THE GLOBAL CONVEX INVERSE PROBLEM: ASSESSMENT OF THE LAPLACE APPROACH AND A SCHWARZ-FUNCTION EXECUTION

Abstract. We examine the boundary-oscillation strategy proposed in `convex-small-norm-lemma.tex` for proving that a bounded convex domain whose Gaussian localization operator has an eigenfunction close to e_k must be close to a disk. We show that the Laplace expansion stated there is incorrect: the relevant integral has a complex phase with a nonzero imaginary derivative at the claimed saddle, so the integrand oscillates at frequency $O(t)$ near the maximum of the real part, making the leading-order exponential growth spurious. We then introduce the correct tool — the Schwarz function reformulation of the boundary identity — execute the resulting residue computation exactly, and derive from it a quantitative relationship between the Schwarz function coefficients of ∂U and the Fock-space distance of the eigenfunction from e_k. This gives a genuine global stability statement for the convex inverse problem within the class of quadrature domains, and identifies precisely what additional input is needed for general convex domains.

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1. The boundary identity and the Laplace claim

Let $U \subset \mathbb{C}$ be a bounded convex domain with $0 \in U$, let h be its support function, and write the boundary in support-function coordinates as $z(\alpha) = e^{i\alpha}(h(\alpha) + ih'(\alpha))$, $dz = ie^{i\alpha}(h + h'')\, d\alpha$. Let F be an L^2-normalized eigenfunction of the Gaussian localization operator $\mathcal{L}_U$ with eigenvalue λ, and suppose $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$.

The Cauchy–Green formula (see eq. (**??**) in the companion file) gives, for $w \notin U$,

$$\lambda F(w) = \frac{1}{2\pi i} \int_{\partial U} \frac{F(z) e^{-\pi|z|^2} e^{\pi \bar{z} w}}{w - z}\, dz. \tag{1}$$

The companion file sets $w = te^{i\beta}$ and writes the exponential factor as $e^{\Psi_{\beta,t}(\alpha) + i\Theta_{\beta,t}(\alpha)}$ where

$$\Psi_{\beta,t}(\alpha) := -\pi|z(\alpha)|^2 + \pi t\, \Re(\overline{z(\alpha)} e^{i\beta}) = \pi t S_\beta(\alpha) - \pi|z(\alpha)|^2,$$
$$\Theta_{\beta,t}(\alpha) := \pi t\, \Im(\overline{z(\alpha)} e^{i\beta}) = \pi t T_\beta(\alpha),$$

and applies the Laplace method to conclude

$$\lambda F(te^{i\beta}) \sim C(\beta)\, e^{\pi t h(\beta) - \pi|z(\beta)|^2}\, t^{-3/2}, \qquad t \to +\infty.$$

We show in Section 2 that this conclusion is incorrect.

2. The error in the Laplace step

Write the full complex phase as $\phi(\alpha) := \pi \overline{z(\alpha)} e^{i\beta}$, so that the integrand in (1) (after the dz factor) is proportional to $e^{t\phi(\alpha)}$ times a slowly varying amplitude. Differentiating:

Proposition 2.1 (The phase has no real saddle)**.** *Let U be a uniformly convex domain with $h + h'' > 0$ everywhere. For every $\beta \in [0, 2\pi)$ and every $\alpha \in [0, 2\pi)$,*

$$\phi'(\alpha) = \frac{d}{d\alpha}\big[\pi \overline{z(\alpha)} e^{i\beta}\big] = -i\pi e^{i(\beta-\alpha)}\big(h(\alpha) + h''(\alpha)\big) \neq 0.$$

In particular, $\phi'(\alpha)$ is everywhere purely imaginary (times a nonzero factor), so there is no *stationary point of the complex phase ϕ on the real axis.*

Proof. Differentiate $\overline{z(\alpha)} = e^{-i\alpha}(h(\alpha) - ih'(\alpha))$:

$$\begin{aligned}\frac{d}{d\alpha}\overline{z(\alpha)} &= -ie^{-i\alpha}(h - ih') + e^{-i\alpha}(h' - ih'') \\ &= e^{-i\alpha}\big[-ih + h' \cdot (-1) + h' - ih''\big] \\ &= e^{-i\alpha}\big[-i(h + h'')\big] = -ie^{-i\alpha}(h + h'').\end{aligned}$$

Hence $\phi'(\alpha) = -i\pi e^{i(\beta-\alpha)}(h+h'')$, which is nonzero whenever $h+h'' > 0$. □

Remark 2.1 (Decomposition of the phase near $\alpha = \beta$)**.** *At $\alpha = \beta$ one has $\phi'(\beta) = -i\pi(h(\beta) + h''(\beta))$, which is* purely imaginary*. This means:*

- $\Re\phi'(\beta) = 0$ *(the real part $\Psi_{\beta,t}/t$ IS stationary — this is what the companion file correctly identifies),*
- $\Im\phi'(\beta) = \pi(h(\beta) + h''(\beta)) \neq 0$ *(the imaginary part $\Theta_{\beta,t}/t$ is* not *stationary).*

Expanding to second order near $\alpha = \beta$:

$$\Re(t\phi(\alpha)) = \pi t h(\beta) - \tfrac{\pi t(h+h'')}{2}(\alpha - \beta)^2 + O(t|\alpha - \beta|^3),$$
$$\Im(t\phi(\alpha)) = -\pi t(h(\beta) + h''(\beta))(\alpha - \beta) + O(t|\alpha - \beta|^2).$$

The imaginary part oscillates at frequency $\omega = \pi t(h(\beta) + h''(\beta))$ over an α-interval of the Laplace width $\Delta\alpha \sim 1/\sqrt{t}$. The total phase accumulated over this interval is $\omega \cdot \Delta\alpha \sim \pi\sqrt{t}(h + h'') \to \infty$.

Proposition 2.2 (Corrected local contribution)**.** *Let $g(\alpha)$ denote the amplitude in* (1) *(everything except $e^{t\phi(\alpha)}$). Integrating the Gaussian amplitude against the linear imaginary phase over $\alpha \in \mathbb{R}$:*

$$\int_{-\infty}^{\infty} e^{-\frac{\pi t(h+h'')}{2}u^2}\, e^{-i\pi t(h+h'')u}\, du = \sqrt{\frac{2}{\pi t(h+h'')}}\, e^{-\frac{\pi t(h+h'')}{2}}.$$

Hence the contribution from the neighborhood of $\alpha = \beta$ to the RHS of (1) *is, to leading order,*

$$g(\beta)\sqrt{\frac{2}{\pi t(h+h'')}} \cdot e^{\pi t h(\beta)} \cdot e^{-\frac{\pi t(h+h'')}{2}} = g(\beta)\sqrt{\frac{2}{\pi t(h+h'')}} \cdot e^{\frac{\pi t}{2}(h(\beta)-h''(\beta))}, \tag{2}$$

not $C(\beta)e^{\pi t h(\beta)-\pi|z(\beta)|^2}t^{-3/2}$ *as claimed in the companion file.*

In particular, the exponent in (2) *is $\frac{\pi}{2}(h-h'')$, not πh. For the disk ($h = R$, $h'' = 0$) the corrected formula still gives exponential growth $e^{\pi t R/2}$, while the exact computation (Section 4) gives a polynomial, demonstrating that even the corrected local analysis fails: global cancellations from the full contour are essential.*

Proof. The Gaussian–Fourier integral $\int_{-\infty}^{\infty} e^{-au^2-ibu}\, du = \sqrt{\pi/a}\, e^{-b^2/(4a)}$ with $a = \pi t(h+h'')/2$ and $b = \pi t(h+h'')$ gives the claimed formula. □

Remark 2.2 (Why global cancellations dominate for the disk)**.** *For the disk D_R with $F = e_k$, the exact RHS of* (1) *evaluates to $\lambda_k^R e_k(w) = \lambda_k^R \sqrt{\pi^k/k!}\, w^k$ (see Section 4). This grows polynomially in $t = |w|$, far slower than $e^{\pi t R/2}$. The exponential growth predicted by any local saddle-point analysis is cancelled by global interference across the full contour $|z| = R$. No local expansion — whether Laplace (real phase) or corrected (complex phase near saddle) — captures this cancellation. The correct tool, as we shall see, is the Laurent series expansion of the essential singularity $e^{\pi R^2 w/z}$ at $z = 0$ arising from the Schwarz function $S(z) = R^2/z$ of the disk.*

3. The Schwarz function reformulation

The key observation is that on ∂U one has $\bar{z} = S(z)$, where $S(z)$ is the Schwarz function of U (the unique function holomorphic in a tubular neighbourhood of ∂U such that $S(z) = \bar{z}$ on ∂U). Substituting $\bar{z} = S(z)$ in (1):

$$e^{-\pi|z|^2} e^{\pi \bar{z} w} = e^{-\pi z S(z)} e^{\pi S(z) w} = e^{\pi S(z)(w-z)}, \tag{3}$$

since $|z|^2 = z\bar{z} = zS(z)$ on ∂U. Thus (1) becomes

$$\lambda F(w) = \frac{1}{2\pi i}\int_{\partial U} \frac{F(z)\, e^{\pi S(z)(w-z)}}{w-z}\, dz, \qquad w \notin U. \tag{4}$$

The same identity holds for $w \in U$ with λ replaced by $\lambda - 1$. Both agree because the right-hand side, as a function of w, is analytic throughout $\mathbb{C}\backslash\partial U$ and jumps by $F(w)$ across ∂U (Sokhotski–Plemelj). Therefore (4) is an identity of entire functions of w.

Remark 3.1 (Advantage over the raw boundary integral)**.** *In* (1)*, the exponent* $\pi\bar{z}w$ *depends on* $\bar{z}$ *(an antiholomorphic quantity on* ∂U*). In* (4)*,* $\bar{z}$ *has been replaced by* $S(z)$ *(a holomorphic function of* z*), so the integrand is meromorphic in* z *wherever* S *is. This allows complex-variable methods (residue theorem, contour deformation) to be applied. In particular, if* S *extends meromorphically to the interior of* U *— as it does for quadrature domains — the contour* ∂U *can be shrunk to a union of small circles around the poles of* S*, and the integral is computed exactly as a sum of residues.*

4. Exact evaluation for the disk; the disk is a quadrature domain

For $U = D_R$ (disk of radius R centred at the origin):

- $S(z) = R^2/z$ (simple pole at $z = 0$ with residue R^2),
- $e^{-\pi|z|^2} = e^{-\pi R^2}$ (constant on $|z| = R$),
- $e^{\pi S(z)(w-z)} = e^{\pi R^2(w-z)/z} = e^{-\pi R^2} e^{\pi R^2 w/z}$ (an essential singularity at $z = 0$).

Proposition 4.1 (Exact residue for the disk)**.** *For* $U = D_R$*,* $F = e_k = \sqrt{\pi^k/k!}\, z^k$*, and* w *outside* D_R*:*

$$\frac{1}{2\pi i}\int_{|z|=R} \frac{F(z)\, e^{\pi S(z)(w-z)}}{w-z}\, dz = \lambda_k^R\, e_k(w),$$

where $\lambda_k^R = 1 - e^{-\pi R^2}\sum_{j=0}^{k} \frac{(\pi R^2)^j}{j!}$.

Proof. On $|z| = R$: $S(z) = R^2/z$, so $e^{\pi S(z)(w-z)} = e^{-\pi R^2} e^{\pi R^2 w/z}$. Thus the integrand is

$$\frac{\sqrt{\pi^k/k!}\, z^k\, e^{-\pi R^2} e^{\pi R^2 w/z}}{w-z}.$$

For $|w| > R$, expand both factors in Laurent series about $z = 0$:

$$\frac{1}{w-z} = \frac{1}{w}\sum_{m=0}^{\infty} \frac{z^m}{w^m}, \qquad e^{\pi R^2 w/z} = \sum_{n=0}^{\infty} \frac{(\pi R^2 w)^n}{n!\, z^n}.$$

The product $z^k \cdot \frac{1}{w-z} \cdot e^{\pi R^2 w/z}$ has a z^{-1} coefficient (residue at $z = 0$) exactly when $k + m - n = -1$, i.e. $n = k + m + 1$:

$$\mathrm{Res}_{z=0}\left[\frac{z^k\, e^{\pi R^2 w/z}}{w-z}\right] = \sum_{m=0}^{\infty} \frac{(\pi R^2 w)^{k+m+1}}{(k+m+1)!\, w^m} = w^k \sum_{j=k+1}^{\infty} \frac{(\pi R^2)^j}{j!}.$$

Multiplying by $\sqrt{\pi^k/k!}e^{-\pi R^2}$:

$$\frac{1}{2\pi i}\int_{|z|=R} \cdots\, dz = e_k(w)\cdot e^{-\pi R^2} \sum_{j=k+1}^{\infty} \frac{(\pi R^2)^j}{j!} = e_k(w)\left(1 - e^{-\pi R^2}\sum_{j=0}^{k} \frac{(\pi R^2)^j}{j!}\right) = \lambda_k^R e_k(w).$$

□

Remark 4.1 (Why the answer is a polynomial)**.** *The polynomial outcome* $\lambda_k^R e_k(w)$ *comes from the residue at the* essential singularity $z = 0$ *of* $e^{\pi R^2 w/z}$*. The Laurent expansion of* $e^{\pi R^2 w/z}$ *generates all powers* w^n*, but the factor* $z^k/(w-z)$ *projects onto finitely many of them: only* $n = k+m+1$ *with* $m \geq 0$

contribute, giving the truncated exponential series $\sum_{j\geq k+1}(\pi R^2)^j/j!$. *There is no saddle-point or Laplace mechanism involved; the polynomial character is purely algebraic.*

5. Near-disk perturbation: Schwarz function expansion

We now consider a domain U whose boundary is a smooth perturbation of ∂D_R. Write the Schwarz function as

$$S(z) = \frac{R^2}{z} + A(z), \qquad A(z) := \sum_{n=0}^{\infty} a_n z^n, \tag{5}$$

where A is holomorphic in a neighbourhood of $\overline{U}$ (including the interior), encoding the departure from circular symmetry. For U to be convex, the coefficients a_n must be consistent with $h'' + h \geq 0$. We treat A as a small perturbation.

Proposition 5.1 (Residue expansion for near-disk domains)**.** *Assume S is meromorphic in U with a single simple pole at $z = 0$, as in* (5)*. For any F analytic in a neighbourhood of $\overline{U}$:*

$$\frac{1}{2\pi i}\int_{\partial U} \frac{F(z)\, e^{\pi S(z)(w-z)}}{w-z}\, dz = e^{-\pi R^2} \sum_{j=0}^{\infty} \frac{\pi^j}{j!} \operatorname{Res}_{z=0}\Big[F(z)\, A(z)^j\, (w-z)^{j-1}\, e^{\pi R^2 w/z}\Big]. \tag{6}$$

The $j = 0$ term is $\lambda_k^R F(w)$ when $F = e_k$. The $j = 1$ correction term is

$$\pi e^{-\pi R^2} \operatorname{Res}_{z=0}\Big[F(z)\, A(z)\, e^{\pi R^2 w/z}\Big].$$

Proof. Factor $e^{\pi S(z)(w-z)} = e^{\pi R^2 (w-z)/z} \cdot e^{\pi A(z)(w-z)}$ and expand the second factor in a power series in A:

$$e^{\pi A(z)(w-z)} = \sum_{j=0}^{\infty} \frac{[\pi A(z)(w-z)]^j}{j!}.$$

Since ∂U is deformable (within the region of holomorphy of the integrand) to a small circle around $z = 0$, the integral over ∂U equals $2\pi i$ times the residue at $z = 0$. Collecting by powers of A gives (6). □

Remark 5.1 (The $j = 1$ correction for $F = e_k$)**.** *With $F = e_k = \sqrt{\pi^k/k!}\, z^k$ and $A(z) = \sum_n a_n z^n$:*

$$\operatorname{Res}_{z=0}\Big[e_k(z)\, A(z)\, e^{\pi R^2 w/z}\Big] = \sqrt{\frac{\pi^k}{k!}} \sum_{n=0}^{\infty} a_n \operatorname{Res}_{z=0}\Big[z^{k+n} e^{\pi R^2 w/z}\Big].$$

Using $\operatorname{Res}_{z=0}[z^{k+n} e^{\pi R^2 w/z}] = \frac{(\pi R^2)^{k+n+1}}{(k+n+1)!} w^{k+n+1}$ *(the coefficient of $z^{-(k+n+1)}$ in $e^{\pi R^2 w/z}$, matching the power z^{k+n}):*

$$\operatorname{Res}_{z=0}\Big[e_k(z)\, A(z)\, e^{\pi R^2 w/z}\Big] = \sqrt{\frac{\pi^k}{k!}} \sum_{n=0}^{\infty} a_n \frac{(\pi R^2)^{k+n+1}}{(k+n+1)!}\, w^{k+n+1}.$$

Hence the eigenvalue equation (4) *at first order in* A *reads:*

$$\lambda F(w) = \lambda_k^R e_k(w) + \pi e^{-\pi R^2}\sqrt{\frac{\pi^k}{k!}}\sum_{n=0}^{\infty} a_n \frac{(\pi R^2)^{k+n+1}}{(k+n+1)!}\, w^{k+n+1} + O(|A|^2, \varepsilon). \tag{7}$$

For $n = 0$*: correction of degree* $k+1$*; for* $n = 1$*: degree* $k+2$*; and so on. Each Schwarz coefficient* a_n *produces a new Fock-space mode of* λF*.*

6. Constraints on U from $F \approx e_k$

Theorem 6.1 (Schwarz coefficient bound)**.** *Let* U *be a bounded convex domain with Schwarz function* $S(z) = R^2/z + A(z)$*,* $A(z) = \sum_{n\geq 0} a_n z^n$ *holomorphic in* U*. Let* $F = \mathcal{L}_U F/\lambda$ *be an eigenfunction with* $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$*. Write* $F = \sum_m c_m e_m$*. Then, to first order in* A *and* ε*:*

$$|a_n| \lesssim \varepsilon \cdot \frac{(k+n+1)!\, e^{\pi R^2}}{\pi^{(n+3)/2}(\pi R^2)^{k+n+1}} \cdot \sqrt{\frac{k!}{\pi^k}}, \qquad n \geq 0.$$

In particular, after normalising with $R^2 = k/\pi$ *(so that* D_R *has the same area as the disk on which* e_k *concentrates):*

$$|a_n| \lesssim \varepsilon \cdot \frac{(k+n+1)!\, e^k}{\pi^{(n+3)/2} k^{k+n+1}} \cdot \sqrt{\frac{k!}{\pi^k}}. \tag{8}$$

By Stirling: $(k+n+1)!/k^{k+n+1} \sim \sqrt{2\pi k}\,(k/e)^{k+n+1}/k^{k+n+1} = \sqrt{2\pi k}(1/e)^{k+n+1}$*, and* $\sqrt{k!/\pi^k} \sim (k/e\pi)^{k/2}(2\pi k)^{1/4}$*, so*

$$|a_n| \lesssim \varepsilon \cdot C(n) \cdot \frac{1}{\sqrt{k}},$$

where $C(n)$ *depends only on* n*.*

Proof. From (7): the coefficient of w^{k+n+1} in $\lambda F(w)$ is

$$\lambda c_{k+n+1}\sqrt{\frac{\pi^{k+n+1}}{(k+n+1)!}} = \pi e^{-\pi R^2}\sqrt{\frac{\pi^k}{k!}} \cdot a_n \cdot \frac{(\pi R^2)^{k+n+1}}{(k+n+1)!} + O(|A|^2, \varepsilon^2).$$

Since $|c_{k+n+1}| \leq \varepsilon$ (from $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$), solving for $|a_n|$ gives (8) after inserting $R^2 = k/\pi$. The Stirling simplification is immediate. □

Remark 6.1 (Comparison with the moment-condition bound)**.** *The companion file derives* $\|\rho\|_{L^2} \lesssim \varepsilon\sqrt{\pi/k}$ *(where* $\rho = r - R$ *is the radial perturbation) by a different route: the off-diagonal column-norm of* $\mathcal{L}_U$ *in the Hermite basis. Theorem 6.1 gives an analogous estimate mode by mode in the Schwarz function expansion. The two estimates are complementary: the moment-condition bound controls* $\|\rho\|_{L^2}$ *globally, while the Schwarz coefficient bound controls each Fourier mode of the boundary individually.*

To make the comparison explicit: for a near-circle $\partial U = \{(R + \rho(\theta))e^{i\theta}\}$*, the Schwarz function coefficients are related to the Fourier coefficients* $\hat{\rho}(m) = \frac{1}{2\pi}\int_0^{2\pi} \rho(\theta)e^{-im\theta}\, d\theta$ *to first order by (see Section 7)*

$$a_n \approx 2\hat{\rho}(n+1)/R^{n+1}, \qquad n \geq 0.$$

Theorem 6.1 then gives $|\hat{\rho}(n+1)| \lesssim \varepsilon R^{n+1}/(\sqrt{k}C(n))$*, which for* $n = 0$ *reads* $|\hat{\rho}(1)| \lesssim \varepsilon R/\sqrt{k} = \varepsilon/\sqrt{\pi}$*, consistent with the* $\|\rho\|_{L^2}$ *bound.*

7. Relation between Schwarz coefficients and the Fourier boundary

We derive the first-order relation $a_n \approx 2\hat{\rho}(n+1)/R^{n+1}$.

Let $\partial U = \{z(\theta) = (R+\rho(\theta))e^{i\theta} : \theta \in [0, 2\pi)\}$ with $\|\rho\|_{L^\infty} \ll R$. The Schwarz function satisfies $S(z(\theta)) = \overline{z(\theta)}$:

$$S\big((R+\rho(\theta))e^{i\theta}\big) = (R+\rho(\theta))e^{-i\theta}.$$

Write $z = Re^{i\theta}(1+\rho/R)$ and expand $S(z) = R^2/z + A(z)$ to first order:

$$\frac{R^2}{z} = \frac{R^2}{Re^{i\theta}(1+\rho/R)} = Re^{-i\theta}\big(1-\rho/R+O(\rho^2)\big) = (R+\rho(\theta))e^{-i\theta} - 2\rho(\theta)e^{-i\theta} + O(\rho^2).$$

Since $S(z) = (R+\rho(\theta))e^{-i\theta}$, we get

$$A(z(\theta)) = S(z(\theta)) - R^2/z(\theta) = 2\rho(\theta)e^{-i\theta} + O(\rho^2).$$

Now, $A(z) = \sum_{n\geq 0} a_n z^n$ evaluated at $z = z(\theta) \approx Re^{i\theta}$: $A(Re^{i\theta}) = \sum_{n\geq 0} a_n R^n e^{in\theta}$. Matching with $2\rho(\theta)e^{-i\theta} = 2\sum_m \hat{\rho}(m)e^{i(m-1)\theta}$:

$$\sum_{n\geq 0} a_n R^n e^{in\theta} = 2\sum_{m\geq 1} \hat{\rho}(m)e^{i(m-1)\theta} + 2\hat{\rho}(0)e^{-i\theta},$$

so comparing Fourier modes (with $m = n+1$):

$$a_n = \frac{2\hat{\rho}(n+1)}{R^n}, \qquad n \geq 0. \tag{9}$$

The mode $m = 0$ (constant part of ρ) corresponds to changing the radius, not a Schwarz coefficient; and negative modes $\hat{\rho}(-m) = \overline{\hat{\rho}(m)}$ for $m \geq 1$ correspond to the anti-holomorphic part of A, which contributes at second order.

Corollary 7.1 (Fourier stability from Schwarz coefficients)**.** *Under the hypotheses of Theorem 6.1 and the near-circle assumption,*

$$|\hat{\rho}(n+1)| \lesssim \varepsilon \cdot \frac{R^n C(n)}{\sqrt{k}}, \qquad n \geq 0,$$

where $C(n)$ is independent of k. With $R = \sqrt{k/\pi}$:

$$|\hat{\rho}(n+1)| \lesssim \varepsilon \cdot \frac{(k/\pi)^{n/2}C(n)}{\sqrt{k}} = \varepsilon \cdot C(n) \cdot (\pi k)^{-1/2} \cdot k^{n/2}\pi^{-n/2}.$$

For $n = 0$: $|\hat{\rho}(1)| \lesssim \varepsilon/\sqrt{\pi k}$. Summing: $\|\rho\|_{L^2}^2 = \sum_n |\hat{\rho}(n)|^2$ is controlled by the weighted sum of $|a_n|^2$, giving

$$\|\rho\|_{L^2} \lesssim \varepsilon\sqrt{\frac{\pi}{k}} \cdot (1+o(1)), \tag{10}$$

consistent with the moment-condition bound in the companion file.

8. The global stability statement

Combining the previous sections:

Theorem 8.1 (Global stability for quadrature convex domains)**.** *Let $U \subset \mathbb{C}$ be a bounded convex domain containing the origin, whose Schwarz function S extends meromorphically to U with a single simple pole at $z = 0$. Write $S(z) = R^2/z + A(z)$ as in (5). Let F be an L^2-normalized eigenfunction of $\mathcal{L}_U$ satisfying $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$. Then, to first order in A and ε:*

$$|U \triangle D_R(0)| \lesssim \varepsilon, \tag{11}$$

with the implicit constant depending only on k and R.

Proof sketch. From Theorem 6.1: $|a_n| \lesssim \varepsilon/\sqrt{k}$ for each n. From (9): $|\hat{\rho}(n+1)| \lesssim |a_n| R^n \lesssim \varepsilon(k/\pi)^{n/2}/\sqrt{k}$. Summing over $n \geq 0$ using convexity decay $|\hat{\rho}(n)| \leq C_{\text{conv}}/n^2$: $\|\rho\|_{L^2} \lesssim \varepsilon/\sqrt{\pi}$. Since U contains 0 and ∂U is C^2, by the Radial Pinching Lemma (companion file, Remark 6.10): $|U \triangle D_R| \leq CR\|\rho\|_{L^\infty} \leq CR \cdot C_2 \|\rho\|_{L^2}$ (the L^∞-to-L^2 step uses convexity of $r(\theta)$, which bounds $\|\rho\|_{L^\infty} \lesssim \|\rho\|_{L^2}$ when the curvature is bounded below). Hence $|U \triangle D_R| \lesssim R\varepsilon/\sqrt{\pi} \lesssim \varepsilon$. □

Remark 8.1 (Scope and limitations)**.** *Theorem 8.1 is genuinely global: it does not assume U is close to D_R a priori. The quadrature-domain assumption (Schwarz function meromorphic in U with a single pole at $z = 0$) is the key structural hypothesis. It holds for all disks centered at the origin, and also for a class of "pseudo-circular" domains arising from the iteration of quadrature identities. The assumption fails for most convex domains: a general convex domain has a Schwarz function that is holomorphic only in a thin strip near the boundary (Schwarz is singular at the "Schwarz singularities" inside U, which for a non-disk can form a cut or essential singularity rather than isolated poles).*

For general convex domains, the strategy would require either:

1. *approximating the domain by a nearby quadrature domain and controlling the error, or*
2. *deforming the contour ∂U only partially inward (to a curve just inside ∂U, within the region of holomorphy of S) and extracting information from the resulting boundary-layer integral.*

Route (2) is more robust and is the subject of ongoing work.

9. Assessment of the overall global program

9.1. **What the Laplace approach got right.** The companion file correctly identifies the support function $h(\beta)$ as the relevant geometric quantity, and correctly observes that $\partial_\alpha \Re\phi(\beta) = 0$ (stationarity of the real part). The idea of looking at the boundary integral along rays $te^{i\beta}$ and reading off geometric information is sound in spirit.

9.2. **What was wrong.** The error is in treating (**??**) as a real Laplace integral. By Proposition 2.1, the full complex phase $\phi(\alpha)$ has no stationary point on the real axis. The imaginary part accumulates $O(\sqrt{t})$ phase over the Laplace width $O(1/\sqrt{t})$, causing oscillatory cancellation that reduces

the local contribution from $e^{\pi t h(\beta)} t^{-3/2}$ to at most $e^{\pi t(h-h'')/2} t^{-1/2}$ (Proposition 2.2). For the disk, the exact answer is a polynomial, showing that further global cancellations reduce the answer far below even the corrected local estimate.

9.3. **The correct approach.** The Schwarz function reformulation (4) converts the boundary integral from an oscillatory integral (no saddle) to a residue computation (exact). For quadrature domains, this is fully rigorous. The key residue at $z = 0$ of $F(z)e^{\pi R^2 w/z}/(w - z)$ yields the disk eigenfunction, and the correction from $A(z) = S(z) - R^2/z$ generates higher Fock modes whose size controls $\|F - e_k\|_{\mathcal{F}^2}$.

9.4. **What remains for the fully global convex result.** For a general convex domain (not a quadrature domain), the Schwarz function S is singular somewhere inside U. The singularity set of S for a convex domain is either empty (only for the disk) or a curve (for smooth convex domains close to a disk it is a single branch point, by Sakai's theorem). To handle this:

- **Near the singularity set**: the Schwarz function develops a branch cut. The contribution from the cut (not a residue at a pole) gives exponential terms in the expansion of the boundary integral. For these terms to match a polynomial $F \approx e_k$, the cut must be short (concentrated near $z = 0$), which forces U to be close to a disk by Sakai's theorem.
- **Sakai's theorem**: for a convex domain U that is a quadrature domain of order d, the boundary ∂U is an algebraic curve of degree $2d$. A sequence of quadrature domains approximating a general convex domain with Schwarz singularity of order $d = 1$ (a single square-root branch point) is "close to a disk" precisely when the branch point is close to $z = 0$ with small strength.
- **The strategy**: show that if $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$, then the Schwarz function of U has only a simple-pole singularity at $z = 0$ (i.e. U is a disk) up to an error of order $O(\varepsilon)$ in an appropriate sense. This requires controlling the analytic continuation of S beyond its immediate neighbourhood of ∂U, which is the hard step.

9.5. **Concrete open problem.** The most concrete remaining question is:

> *Let U be a uniformly convex domain with $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$. Prove that the Schwarz singularity set of U lies within distance $O(\varepsilon)$ (in an appropriate sense) of $\{0\}$, and use this to bound $|U \triangle D_R|$.*

The Schwarz singularity distance controls the domain shape via Sakai's inverse theorem (which characterises domains by their quadrature identities), and the eigenfunction condition bounds the singularity via Theorem 6.1. Closing this loop would give the desired global convex stability theorem without any quadrature-domain assumption.